

Derivation of the Product rule

In today's session³¹ we are going to learn where the product and quotient rules for derivatives come from.

³¹ February 12

Starting with the product rule, we recall that $f(x) = u(x)v(x)$. Then, by using the definition of the derivative, we get

$$\frac{df(x)}{dx} = \lim_{h \rightarrow 0} \frac{u(x+h)v(x+h) - u(x)v(x)}{h}.$$

In order to get somewhere, let's add a clever 0, $u(x+h)v(x) - u(x+h)v(x)$,

$$\frac{df(x)}{dx} = \lim_{h \rightarrow 0} \frac{u(x+h)v(x+h) - u(x+h)v(x) - u(x)v(x) + u(x+h)v(x)}{h}.$$

This clever 0 allow us to factor terms,

$$\frac{df(x)}{dx} = \lim_{h \rightarrow 0} \left[u(x+h) \frac{v(x+h) - v(x)}{h} + v(x) \frac{u(x+h) - u(x)}{h} \right].$$

By applying the limit and using properties,

$$\begin{aligned} \lim_{h \rightarrow 0} \left[u(x+h) \frac{v(x+h) - v(x)}{h} \right] &= \left[\lim_{h \rightarrow 0} u(x+h) \right] \left[\lim_{h \rightarrow 0} \frac{v(x+h) - v(x)}{h} \right] \\ &= u(x) \frac{dv(x)}{dx} \end{aligned}$$

$$\begin{aligned} \lim_{h \rightarrow 0} \left[v(x) \frac{u(x+h) - u(x)}{h} \right] &= \left[\lim_{h \rightarrow 0} v(x+h) \right] \left[\lim_{h \rightarrow 0} \frac{u(x+h) - u(x)}{h} \right] \\ &= v(x) \frac{du(x)}{dx} \end{aligned}$$

We show that,

$$\frac{df(x)}{dx} = \lim_{h \rightarrow 0} \frac{u(x+h)v(x+h) - u(x)v(x)}{h} = u(x) \frac{dv(x)}{dx} + v(x) \frac{du(x)}{dx}.$$

Class activities

Compute the derivatives using the product, quotient and power rules of the following functions. You might need the rules of transcendental functions.

$$\begin{aligned} f(x) &= \frac{\ln(x)}{e^x} \\ g(x) &= e^x \ln(x) \sin(x) \\ h(x) &= \ln(x) \sin(x) \end{aligned}$$